Model Interpretability

Feature Importance

What we did

- We used the **Random Forest model's feature importance scores** to understand which variables most influenced loan approval decisions.

Key Findings

- **Credit Score** is the most influential feature. Applicants with higher credit scores are significantly more likely to be approved.
- **Debt-to-Income Ratio (DTI)** is a strong negative indicator. As DTI increases, approval probability decreases.
- **Previous Defaults** heavily reduce approval chances, signaling high credit risk.
- **Income** positively impacts approval but is less influential than credit behavior variables.

Business Interpretation

The model prioritizes creditworthiness and repayment capacity over demographic attributes, aligning well with standard banking risk policies.

Partial Dependence Plots (PDP) – Interpretation

Credit Score

- Approval probability increases sharply once credit score crosses approximately **650**.
- Scores above **720** show diminishing returns, indicating saturation in approval likelihood.

Debt-to-Income Ratio

- Approval probability declines rapidly when DTI exceeds **40%**.
- Applicants with DTI above **50%** are almost always declined.

Income

- Higher income improves approval probability, but the effect is gradual.
- Income alone cannot compensate for poor credit history.

Summary

The model behaves intuitively: strong credit history and manageable debt matter more than income alone.

Final Model Insights

- The model mirrors real-world underwriting logic.
- Credit behavior outweighs demographic factors.
- High-risk applicants are clearly identifiable.
- The Random Forest model provides strong predictive performance while remaining explainable.